CAPSTONE PROJECT

WEEK 1 – DESIGN DOCUMENT

PS-055: Recruitment ATS with Interview Scheduling
and Offer Approval Matrix

Team ID	T058
Member 1	Baby V S – 1AT24CG012
Member 2	Bhagyashree Rama Naik – 1AT24CG013
Section	Computer Science and Design (CSD)
Domain	Enterprise / HR – Recruitment
Selected Language	**Java 17**
Checkpoint	Week 1 – Design

Prepared for the 10 SECONDS Capstone Project

Design deliverables: glossary, class diagram, state diagram and business-rule register

1. Purpose and Scope

This document defines the design for PS-055, a command-file-driven back-end engine for an IT services company's recruitment process. It manages requisitions and candidate applications, interview scheduling, offer calculations, approval routing, offer outcomes and pipeline reporting. The system has no graphical user interface; it reads supplied input files and prints deterministic results.

In scope: requisition and candidate registration, application stages, interview panels and availability, CTC and bonus calculations, approval matrix and SLA, headcount reservation, offer validity, requisition ageing, and date-based pipeline reports.

Selected implementation language: Java 17. The project will use Java 17 and object-oriented design. Suggested build and test tools are Maven and JUnit 5; the final implementation should use one consistent project setup.

2. Glossary

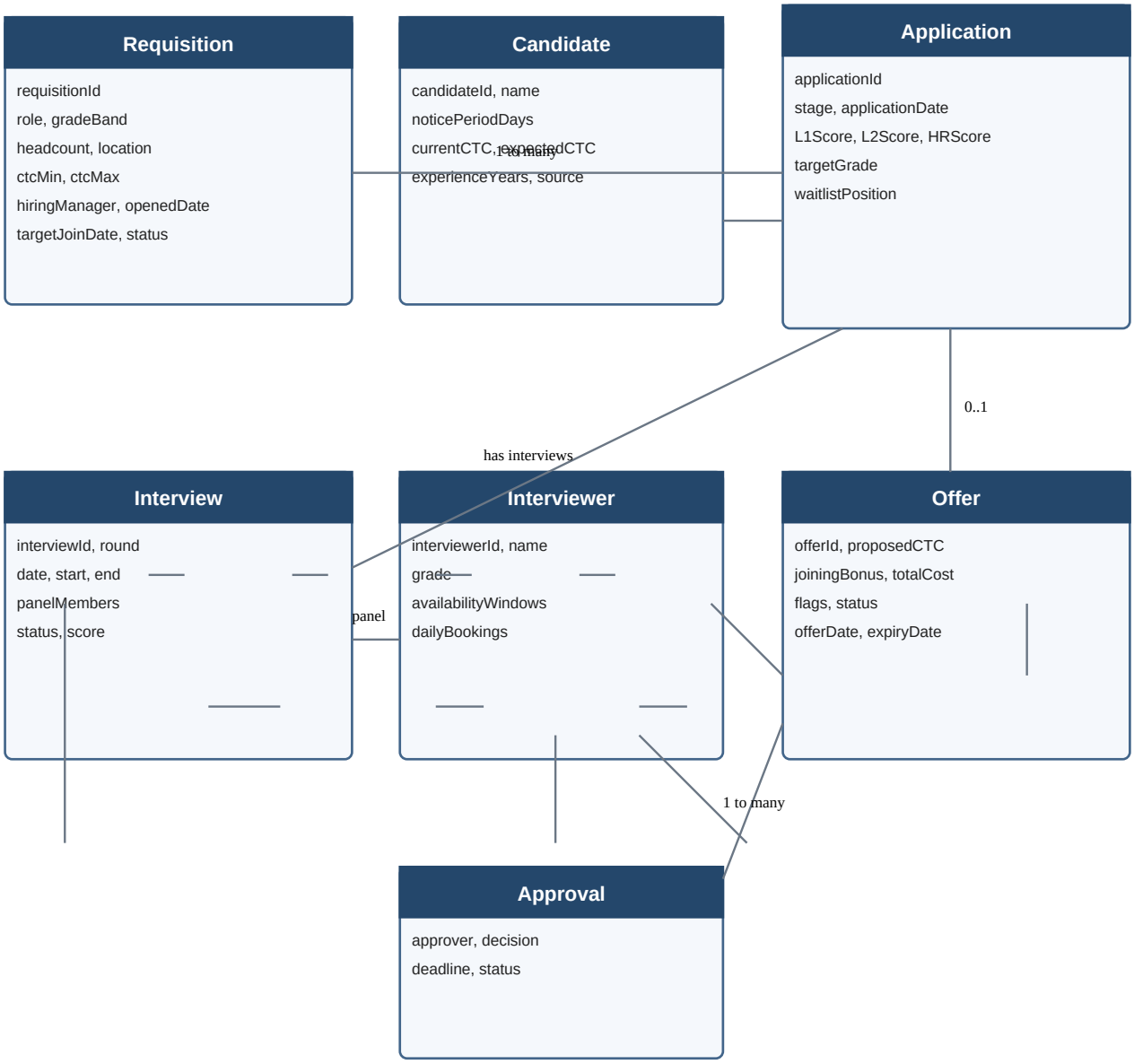
Term	Definition
ATS	Applicant Tracking System used to manage candidates and recruitment activities.
Requisition	A request to hire one or more employees for a defined role.
Candidate	A person whose details are registered in the recruitment system.
Application	A candidate's application to a particular requisition; it owns the stage and interview scores.
CTC	Cost to Company; the annual compensation amount proposed for a candidate.
Grade band	A job level from G1 to G6; grade comparisons determine L2 panel eligibility.
Headcount	Number of positions approved for a requisition.
Panel interview	An interview attended by one or more assigned interviewers.
Availability window	A date and time interval during which an interviewer can attend an interview.
State machine	A model defining valid states and permitted transitions.
Business rule	A mandatory condition the system validates before allowing an action.
SLA	Service-level agreement; the allowed time for an approver to act.
Approval matrix	Rules that determine required approvers and their order.
OUT_OF_BAND	An offer flag raised when expected CTC exceeds band maximum by more than 10%.
WAITLIST	A queue for candidates who cannot reserve requisition headcount yet.
AT_RISK	A requisition open for more than 45 days.
NO_SLOT	Scheduling result when no valid interview slot is found in the search period.
Command-file driven	The application processes structured commands from a file, not a graphical UI.

3. System Components

Component	Responsibility
CommandProcessor	Reads commands and validates syntax; invokes services.
RecruitmentService	Registers requisitions, candidates and applications.
StageTransitionService	Validates transitions and returns reason codes for illegal transitions.
InterviewScheduler	Finds earliest valid slots and checks availability, duration, load, gaps and panel grade.
OfferCalculator	Calculates proposed CTC, bonus, referral/agency cost and total cost.
ApprovalService	Builds sequential approval chain, records decisions, deadlines and escalation.
HeadcountService	Reserves/releases positions and orders waitlisted applications.
LifecycleService	Handles offer expiry, acceptance, decline, joining and requisition ageing.
PipelineReportService	Reports stage counts, pending approvals and at-risk requisitions.
HolidayCalendar	Stores supplied holidays and supports working-day calculations.

4. Class Diagram

The model separates a person's candidate record from an application to a requisition. Each application maintains its own stage, scores, interviews and offer lifecycle.



Supporting classes/services: BusinessRuleEngine, OfferCalculator, InterviewScheduler, ApprovalService, HeadcountService, HolidayCalendar, CommandProcessor and PipelineReportService. These may be implemented as Java classes within appropriate packages.

Relationships: one requisition has many applications; each application belongs to one candidate and one requisition; an application can have multiple interviews and at most one active offer; interviews have one or more interviewers; offers contain ordered approval records.

5. State Diagrams

5.1 Application state machine

Primary path: SCREENED → L1 → L2 → HR → OFFER_DRAFT → OFFER_APPROVED → OFFERED → ACCEPTED → JOINED. An OFFERED candidate may DECLINE. An unaccepted offer becomes WITHDRAWN on day 8. A score below 2.0 sends the application to REJECTED immediately. WAITLIST and ON_HOLD are additional application states. NO_SLOT is a scheduling outcome and does not change the stage.

Transition guards: L2 requires L1 score ≥ 3.0; HR requires L2 score ≥ 3.5. Scores exactly 3.0 and 3.5 pass. Any score < 2.0 causes immediate rejection. Illegal transitions return a documented reason code.

5.2 Requisition states

State	Meaning
OPEN	Accepts applications and offer reservations while headcount is available.
AT_RISK	Open for more than 45 days; may still be active.
CLOSED	Closed when headcount is filled under the implemented rule, or auto-closed after more than 90 days if no headcount is reserved.

6. Business-Rule Register

The register converts the project brief into traceable, testable rules. Use the BR identifier in source comments where useful, unit-test names and the decision log.

Rule ID	Rule name	Requirement
BR-01	Requisition registration	Require role, grade band G1–G6, headcount, location, CTC band min/max, hiring manager and opened date. Validate min ≤ max and positive headcount.
BR-02	Candidate registration	Require notice period, current CTC, expected CTC, experience and source REFERRAL / PORTAL / AGENCY / INTERNAL.
BR-03	Application registration	Link candidate to a valid requisition; store application date and target grade.
BR-04	Legal transitions	Allow only defined transitions; reject illegal transitions with a stable reason code.
BR-05	L1-to-L2 gate	L2 requires L1 score ≥ 3.0/5; exactly 3.0 passes.
BR-06	L2-to-HR gate	HR requires L2 score ≥ 3.5/5; exactly 3.5 passes.
BR-07	Immediate rejection	A score < 2.0 in any round moves the application to REJECTED immediately.
BR-08	Interview durations	L1 = 60 minutes; L2 = 45 minutes; HR = 30 minutes.
BR-09	Daily interviewer limit	An interviewer may attend at most 3 interviews per day.
BR-10	Inter-interview gap	Require at least 15 minutes between consecutive interviews for the same interviewer; 14 minutes is invalid.
BR-11	L2 panel grade	The L2 panel must include an interviewer graded ≥ target grade + 1.
BR-12	Availability and slot search	Choose the earliest valid slot in the next 10 working days, Mon–Fri, excluding supplied holidays, that fits all panel members.
BR-13	Window boundary	A slot ending exactly at a window's end is valid if it otherwise fits; overlap checks must be consistent.
BR-14	No slot result	If no valid slot exists, print NO_SLOT and keep the application in its current stage.
BR-15	Offer CTC calculation	Start from expected CTC; apply the 1.40 × current CTC cap and band maximum cap; enforce band minimum floor. Document deterministic precedence if floor and cap conflict.
BR-16	OUT_OF_BAND flag	If original expected CTC > band maximum × 1.10, flag OUT_OF_BAND. Exactly 10% above is not flagged; 10.01% above is.
BR-17	VP before drafting	An OUT_OF_BAND offer cannot be drafted without VP approval.
BR-18	Joining bonus – quick join	■50,000 if notice period ≤ 30 days and candidate joins within 45 days of offer, except INTERNAL.
BR-19	Joining bonus – buyout	■1,00,000 if notice period > 60 days and candidate buys out notice, except INTERNAL. Reimbursement cap is one month of current CTC / 12.
BR-20	Internal bonus exemption	Joining bonus is zero for INTERNAL candidates.
BR-21	Referral fee	Pay ■25,000 at JOINED + 90 days for REFERRAL candidates who remain JOINED for 90 days.
BR-22	Agency fee	Agency fee is 8.33% of offered CTC and payable only if candidate remains JOINED for 90 days.
BR-23	Total-cost cap	If offer + applicable fee + bonus > band maximum × 1.15, append CFO approval.
BR-24	Approval – within band	Offers ≤ band maximum require Hiring Manager approval, unless another rule adds an approver.
BR-25	Approval – up to 110%	Offers > band maximum and ≤ band maximum × 1.10 require Hiring Manager then HR Head, unless another rule adds an approver.
BR-26	Approval – above 110%	Offers > band maximum × 1.10 or flagged OUT_OF_BAND require Hiring Manager, HR Head and VP in order.
BR-27	Sequential approval	Approvals are sequential; any REJECT ends the chain.
BR-28	Approval SLA	Each approver has two working days. On breach, record TIMED_OUT and auto-escalate to next approver except VP and CFO are never auto-skipped; exclude weekends and holidays.
BR-29	Offer validity	Accept through day 7 at 23:59; on day 8 at 00:00 an unaccepted offer becomes WITHDRAWN.
BR-30	Declined offer	A DECLINED offer releases reserved headcount immediately.
BR-31	Headcount reservation	OFFER_DRAFT reserves one free headcount; no new draft if no headcount is free.
BR-32	Waitlist ordering	Order WAITLIST by L2 score descending, then application date ascending.
BR-33	Last headcount concurrency	When candidates compete for the last position, apply the waitlist order atomically; score ties use application date.
BR-34	Requisition ageing	More than 45 days open → AT_RISK. More than 90 days → auto-close if no headcount is reserved.
BR-35	Close requisition	When a requisition closes, move all non-OFFERED applications to ON_HOLD.
BR-36	Notice-period risk	If notice period > 90 days, offer only if target join date is at least notice-period days after offer date.
BR-37	Pipeline report	For a requested date, print candidate counts per stage, offers pending per approver and requisitions at risk.
BR-38	Command processing	Read command files and output deterministic transitions, slots, offer calculations, approvals and final candidate statuses.

7. Decision Order and Workflows

7.1 Offer decision order

Use one documented sequence so all outputs and tests apply the same logic. Clarify ambiguous cases with the mentor where possible.

Step	Decision
1. Eligibility	Validate stage gates, notice-period restriction, requisition status and headcount.
2. Preliminary CTC	Start at expected CTC; apply the 1.40 × current CTC cap and band maximum cap; enforce band minimum floor. Record the chosen precedence if floor conflicts with a cap.
3. OUT_OF_BAND	Compare original expected CTC to band maximum × 1.10 before applying the offer cap.
4. Bonus and fees	Calculate joining bonus and applicable referral/agency cost using eligibility and payment timing.
5. Total cost	Compute offered CTC + applicable fee + bonus. If it exceeds band maximum × 1.15, require CFO approval.
6. Approval chain	Hiring Manager; add HR Head for offer above band max through 110%; add VP above 110% or for OUT_OF_BAND; append CFO if total-cost rule fires.
7. Decisions	Record approver, decision and deadline. Stop on rejection; apply timeout rules only where permitted.
8. Draft and offer	Do not draft OUT_OF_BAND offers until VP approval. Reserve headcount at OFFER_DRAFT as specified.

7.2 Scheduling order

Determine round duration; enumerate candidate slots chronologically across the next 10 working days; exclude weekends and holidays; check all panel members' availability; check conflicts, 3-interview daily limit and 15-minute gap; validate L2 panel grade; select the first valid slot. If none exists, print NO_SLOT and leave the stage unchanged.

8. Edge Cases and Validation Checklist

Test case	Expected behaviour
L1 score exactly 3.0	Passes L2 gate.
L2 score exactly 3.5	Passes HR gate.
Any score < 2.0	Immediately REJECTED.
Slot ends at window end	Valid if all other constraints pass.
14-minute gap	Invalid; minimum gap is 15 minutes.
Expected CTC exactly 110% of band max	Not OUT_OF_BAND.
Expected CTC 110.01% of band max	OUT_OF_BAND.
Accept at day 7, 23:59	Valid if other conditions pass.
Accept at day 8, 00:00	Expired; status WITHDRAWN.
Friday SLA crossing weekend/holiday	Count working days only, excluding supplied holidays.
Two candidates for last headcount	L2 score descending, then application date; reserve atomically.
INTERNAL, notice 0, expected CTC below current CTC	Bonus zero; CTC follows documented cap/floor precedence.
No valid slot in 10 working days	NO_SLOT; current stage unchanged.
VP/CFO SLA breach	Do not auto-skip; apply explicit exception handling.

9. Assumptions and Decision Log

The brief contains details that should be explicit before implementation. Proposed decisions are listed below; confirm with the mentor if possible.

Decision ID	Topic	Proposed decision / clarification
DEC-01	OUT_OF_BAND	Compare original expected CTC to band maximum × 1.10; exactly 110% is not flagged.
DEC-02	CTC floor/cap conflict	If band minimum conflicts with the current-CTC cap or band maximum, use a deterministic documented precedence and validate requisition bands.
DEC-03	Interview intervals	An interview may end at the window boundary. Conflict checks and the separate 15-minute gap rule must both pass.
DEC-04	Offer validity	Acceptance through day 7 at 23:59 is valid; day 8 at 00:00 is expired.
DEC-05	Approval timeout	Working days exclude weekends and supplied holidays. VP and CFO are not auto-skipped.

Decision ID	Topic	Proposed decision / clarification
DEC-06	Total cost	Use offered CTC + applicable fee + bonus; confirm whether deferred referral cost is included at approval time.
DEC-07	Concurrent headcount	Process competing drafts deterministically and commit reservations atomically.
DEC-08	State model	Keep application state, offer status and requisition status separate.

10. Initial Test Strategy

Use unit tests for each business rule and integration tests for full command-file scenarios. Each test should record BR ID, input, expected output and actual result. Include positive, negative and boundary tests.

Test group	Coverage
Domain validation	Missing fields, invalid grade, invalid headcount, invalid source, min CTC > max.
State transitions	Legal/illegal transitions; exact score thresholds; score < 2.0 rejection.
Scheduler	Window boundary, overlaps, 14-minute gap, daily cap, L2 grade, weekends and holidays.
Offer calculator	CTC caps/floor, 110% threshold, bonus eligibility, INTERNAL exemption, agency/referral fee.
Approvals	All matrix tiers, CFO append, rejection, working-day SLA, VP/CFO no-skip.
Lifecycle/headcount	Day 7/day 8, decline release, final headcount, waitlist tie-break, requisition ageing.
Reports	Stage counts, pending approvers and requisition risk for given dates.

11. Java 17 Project Setup Recommendation

Suggested structure for the repository (Maven is recommended but not mandatory):

```
T058-PS-055/
pom.xml
README.md
docs/
week1-design.pdf
decision-log.md
src/main/java/...
model/
service/
rules/
io/
src/test/java/...
```

Use Java enums for fixed values such as candidate stages, candidate sources, interview rounds, approval decisions and statuses. Use java.time types (LocalDate, LocalDateTime, LocalTime) for dates and scheduling. Use BigDecimal for CTC and money calculations to avoid floating-point rounding issues. Use JUnit 5 for unit tests.

12. Week 1 Completion Checklist

■ Glossary prepared.

■ Class diagram and relationships prepared.

■ Application and requisition state diagrams prepared.

■ Business-rule register BR-01 to BR-38 prepared.

■ Edge cases and boundary conditions listed.

■ Offer and scheduling decision order documented.

■ Assumptions and decision log prepared.

■ Selected language recorded as Java 17.

End of Week 1 Design Document — Team T058 — PS-055 — Java 17